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| Exp No: 11 | Routing at Network Layer - Static Routing and RIP |
|---|---|

## AIM:

To simulate static routing and the RIP dynamic routing protocol configuration using Cisco Packet Tracer.

## ALGORITHM / PROCEDURE:

### (a) Static routing

1. Place 3 routers (2911 with HWIC-2T serial card) and 2 PCs. Connect PC0 to R1 and PC1 to R3 with a crossover cable, and connect the routers using **serial DCE** cables (R1-R2 and R2-R3).
2. Assign IP addresses to all the router interfaces and the PCs as shown in the address table. Set the default gateways (`192.168.1.1` and `192.168.2.1`).
3. On the DCE side of each serial link give `clock rate 64000` and use `no shutdown` on all interfaces.
4. Ping from PC0 to PC1. It fails because the routers do not know the remote networks.
5. On each router add a static route for every network that is not directly connected: `ip route <network> <mask> <next-hop>`
6. Verify with `show ip route` and ping PC0 to PC1 again. Use `tracert` to see the path.

### (b) RIP (Routing Information Protocol)

1. Remove the static routes using `no ip route ...` (or build the same topology again).
2. On every router enter RIP configuration mode with `router rip` and `version 2`.
3. Advertise the directly connected networks using the `network <address>` command and disable auto summary using `no auto-summary`.
4. Wait for about 30 seconds (RIP update timer). Check the learnt routes with `show ip route` and `show ip protocols`.
5. Ping PC0 to PC1 and trace the path.

| Device | Interface | IP address | Subnet mask |
|---|---|---|---|
| R1 | Gig0/0 | `192.168.1.1` | `255.255.255.0` |
| R1 | Se0/0/0 (DCE) | `10.0.0.1` | `255.255.255.252` |
| R2 | Se0/0/0 | `10.0.0.2` | `255.255.255.252` |
| R2 | Se0/0/1 (DCE) | `10.0.0.5` | `255.255.255.252` |
| R3 | Se0/0/1 | `10.0.0.6` | `255.255.255.252` |
| R3 | Gig0/0 | `192.168.2.1` | `255.255.255.0` |
| PC0 | NIC | `192.168.1.2` | `255.255.255.0` |
| PC1 | NIC | `192.168.2.2` | `255.255.255.0` |

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## Commands used

| Static routing | RIP |
|---|---|
| `R1(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2` | `R1(config)#router rip` |
| `R1(config)#ip route 10.0.0.4 255.255.255.252 10.0.0.2` | `R1(config-router)#version 2` |
| `R2(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1` | `R1(config-router)#network 192.168.1.0` |
| `R2(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.6` | `R1(config-router)#network 10.0.0.0` |
| `R3(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.5` | `R1(config-router)#no auto-summary` |
| `R3(config)#ip route 10.0.0.0 255.255.255.252 10.0.0.5` | `(similar for R2: 10.0.0.0 and R3: 192.168.2.0)` |

# DIAGRAM:

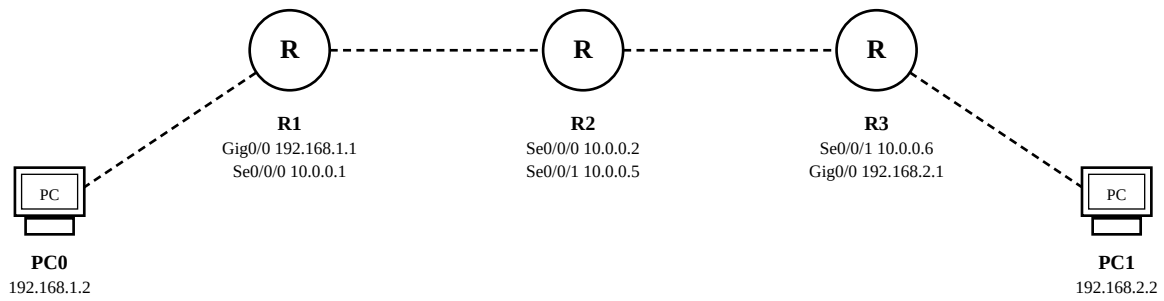
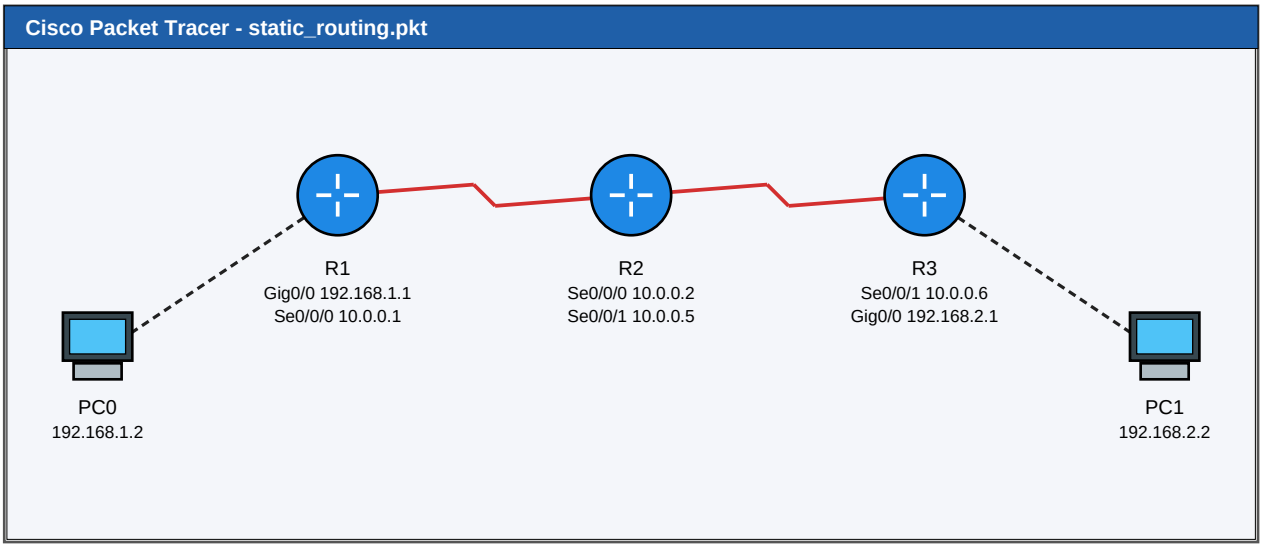



**Fig 1: Three routers connected by serial links (dotted lines are serial links)**

# OUTPUT:

## Part (a) - Static routing




**Packet Tracer view (red lightning lines are serial links)**

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**PC0 - Command Prompt (before static routes)**

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

**R1 - CLI**

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 2 subnets
C       10.0.0.0 is directly connected, Serial0/0/0
S       10.0.0.4 [1/0] via 10.0.0.2
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
S    192.168.2.0/24 [1/0] via 10.0.0.2
```

**PC0 - Command Prompt (after static routes)**

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=3ms TTL=125
Reply from 192.168.2.2: bytes=32 time=2ms TTL=125
Reply from 192.168.2.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),

C:\>tracert 192.168.2.2

Tracing route to 192.168.2.2 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.1.1
  2   1 ms      1 ms      0 ms      10.0.0.2
  3   2 ms      1 ms      2 ms      10.0.0.6
  4   2 ms      2 ms      1 ms      192.168.2.2

Trace complete.
```

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## Part (b) - RIP

**R1 - CLI**

```
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 192.168.1.0
R1(config-router)#network 10.0.0.0
R1(config-router)#no auto-summary
R1(config-router)#end
R1#show ip route rip
     10.0.0.0/30 is subnetted, 2 subnets
R       10.0.0.4 [120/1] via 10.0.0.2, 00:00:12, Serial0/0/0
R    192.168.2.0/24 [120/2] via 10.0.0.2, 00:00:12, Serial0/0/0
R1#show ip protocols
Routing Protocol is "rip"
 Sending updates every 30 seconds, next due in 17 seconds
 Invalid after 180 seconds, hold down 180, flushed after 240
 Outgoing update filter list for all interfaces is not set
 Incoming update filter list for all interfaces is not set
 Redistributing: rip
 Default version control: send version 2, receive 2
 Automatic network summarization is not in effect
 Maximum path: 4
 Routing for Networks:
   10.0.0.0
   192.168.1.0
 Routing Information Sources:
   Gateway         Distance      Last Update
   10.0.0.2             120      00:00:12
 Distance: (default is 120)
```

**PC0 - Command Prompt**

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=2ms TTL=125
Reply from 192.168.2.2: bytes=32 time=2ms TTL=125
Reply from 192.168.2.2: bytes=32 time=1ms TTL=125
Reply from 192.168.2.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

| Fire | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num |
|---|---|---|---|---|---|---|---|---|
| | Successful | PC0 | R1 | ICMP | | 0.000 | N | 0 |
| | Successful | R1 | R2 | ICMP | | 0.001 | N | 1 |
| | Successful | R2 | R3 | ICMP | | 0.002 | N | 2 |
| | Successful | R3 | PC1 | ICMP | | 0.003 | N | 3 |
| | Successful | R1 | R2 | RIP | | 30.000 | N | 4 |
| | Successful | R2 | R3 | RIP | | 30.001 | N | 5 |

| Point | Static routing | RIP |
|---|---|---|
| Route entry | Typed manually by the admin (S) | Learnt automatically from neighbours (R) |
| Administrative distance | 1 | 120 |
| Updates | None | Every 30 seconds |
| Link failure | Admin has to change the route | Finds another path automatically |
| Metric | Not used | Hop count (max 15) |
| Best for | Small and stable networks | Small to medium networks |

## RESULT:

Thus static routing and RIP were configured and simulated in Cisco Packet Tracer. Ping between the two remote networks failed without routes and became successful after configuring the static routes and also after configuring RIP.